

Business Requirements Document (BRD)

1. Overview

This document outlines the functional and technical requirements for the development of a Mixer Batch Reporting Software. The goal is to facilitate the creation, execution, and reporting of ingredient-based mixing batches with robust user management, traceability, and analytics.

2. User Roles & Access Control

- Admin: Full access to all modules. Can manage users, materials, recipes, reports.
- Operator: Limited to batch execution and viewing reports. Cannot modify recipes or materials.

Login is required for all users. Authentication will be handled via secure credentials with optional 2FA (future enhancement).

3. Home Screen (Dashboard)

Top Bar: App Logo, App Name, Logged-in User (Operator/Admin), Real-Time Clock

Navigation Buttons: [Create Recipe], [Material Management], [Run Batch], [Reports]

Status Panel: Active Batches Count, Current Equipment Status

4. Material Management Screen

Purpose: Create, edit, and manage material records.

Material Info: Material Name (Text Field), Material Code (Alphanumeric), [Save] Button

Data Grid: Columns – Material name, material code, timestamp, Created by username

Controls: [Add], [Delete], [Refresh]

Notes:

- Material Code must be unique.

5. Recipe Management Screen

Purpose: Define multi-step recipes using registered materials.

Recipe Info: Recipe Name (Text Field), Created By (Auto-fill), Created On (Auto Timestamp), [Save Recipe]

Steps Grid: Ingredient (Dropdown), Setpoint (Numeric Input), Equipment (Dropdown), Step Order, Tolerance (Numeric %)

Controls: [Add Step], [Delete Step], [Reorder Steps]

Side Panel: List of Existing Recipes with [Edit], [Delete] buttons

Notes:

- Steps must be executed in the defined order.
- Tolerance determines acceptable variance during batch execution.
- Equipment dropdown is pre-populated from master material list.

6. Run Batch Screen

Purpose: Execute batches based on predefined recipes.

Batch Info: Recipe (Dropdown), Operator (Auto-fill), [Start Batch]

Active Step View: Display current Ingredient, Setpoint, Equipment

QR Scan Section: QR Input Field, Actual Value Input, [Log Entry]

History Table: Timestamp, Scanned QR Code, Entered Actual Value

Controls: [End Batch], [Print Report], [Back]

As the steps executes during recipe running, material should be highlighted what material should be taken next and wait till the step finishes.

Validation:

- QR must match expected material and step.
- Actual value within tolerance range is highlighted (green); out-of-range flagged (red) and weighing acknowledge button should get highlighted and print QR code button will be visible so that QR can be previewed and printed.

*** actual value from Load cell should be integrated in software application. We have to look after ways to take actual values into software screen.

7. QR Code Generator

Purpose: Generate QR labels for materials.

Input: Material Code, Setpoint, Actual Value, Material Name, Equipment

Output: QR Code Generation Preview

Controls: [Generate], [Print]

QR Format Example: 11000012,580.25,498.25,Material-X,Equipment-A



8. Reports Module

Purpose: View and export batch execution history.

Filters: Batch ID, Date Range, Recipe Name

Results Table: Batch ID, Recipe Name, Operator, Start/End Time, [Export]

Detail Panel: Step-by-step info: Ingredient, Setpoint, Actual Value, Equipment, Timestamp

Controls: [Export to Excel], [Export to PDF]

9. Analytics & Visualization

Purpose: Visual comparison of Setpoint vs Actual values.

Visualization: Bar Chart

X-Axis: Step Number / Ingredient

Y-Axis: Value (kg/ml)

Bars: Setpoint vs Actual (Side-by-Side)

10. Non-Functional Requirements

Platform: Web-based (Desktop), optional tablet interface

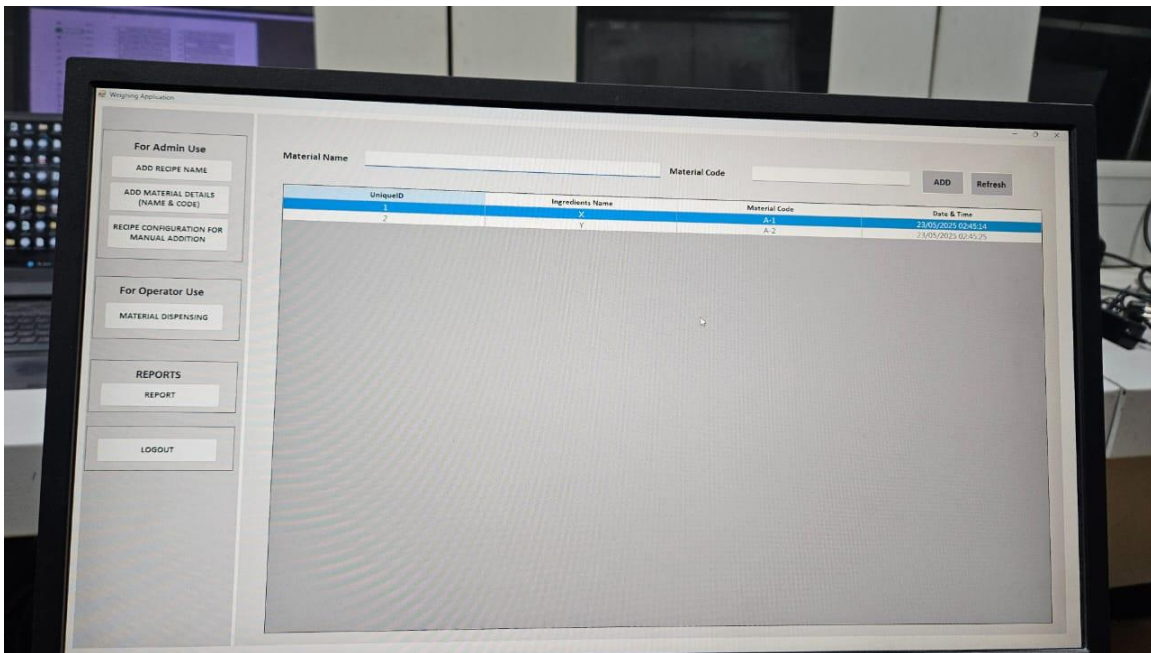
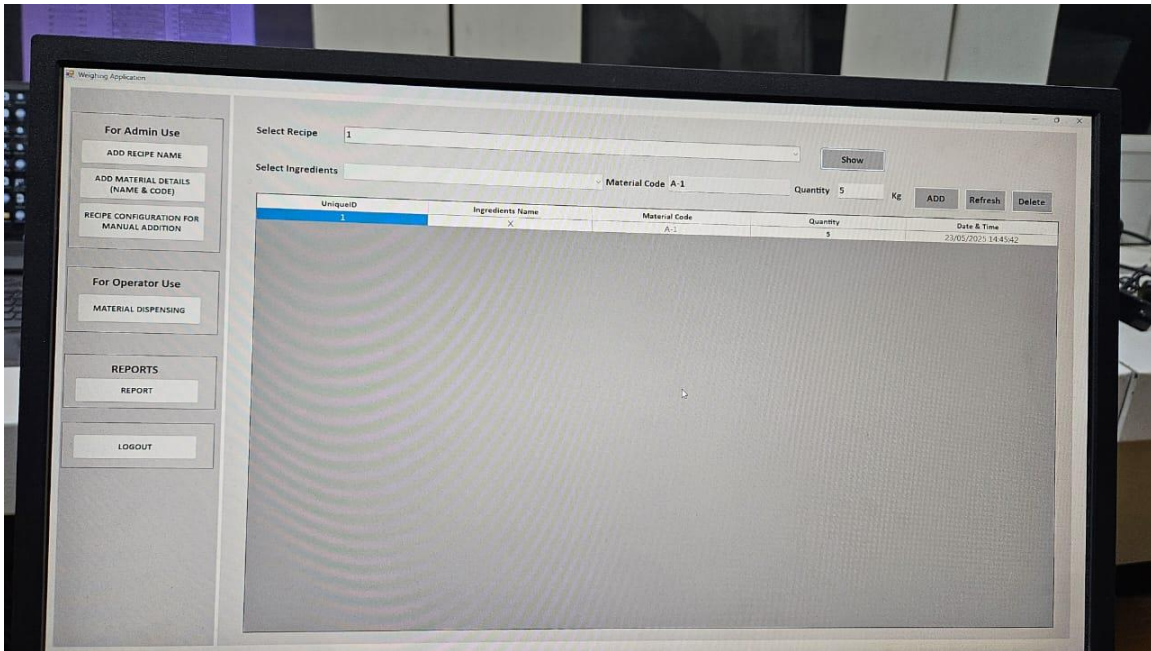
Technology Stack: To be proposed by development team

Data Storage: Secure, encrypted; option for cloud or on-premise DB

Backup: Daily automated backups

11. Deliverables

- Complete software with source code
- Admin and operator user manuals
- Deployment guide and training session
- Support and maintenance contract (optional)



For Admin Use

ADD RECIPE NAME

ADD MATERIAL DETAILS
(NAME & CODE)

RECIPE CONFIGURATION FOR
MANUAL ADDITION

For Operator Use

MATERIAL DISPENSING

REPORTS

REPORT

LOGOUT

BatchNo1

Select Recipe Name1

Batch Size5000 KI

Select ShiftB

Select IngredientsNone

MatCode0

Weight SetPoint

ID	BarCode	BatchNo	Ingredients	SPval	PIval	Weighing Status	TS
1	0000000010000000A-1001...	1	X	5	5	OK	06/06/2025 18:11:00

Ingredient Weight

Actual Weight

Weighing Ack

Records

0000000010000000A-1001K00500

MatCodeA-1

BatchNo1

MatName5000 KI

BatchSize5000 KI

ShiftB

DateTime06/Jun/2025 18:11:00

NetWt5

Print